The (λ, N) sweep on ring count

`marginal` against `edlm`, MDLM and UDLM

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1 Setup

252 runs, 1,024 samples each at $T = 32$ steps, seed 1, reward `ring_count`. Grid: $\lambda \in \{0, 0.5, 1, 2, 5, 10, 30, 100, 1000\}$ against $N \in \{10, 30, 100, 300, 500, 1000, 2000\}$, for both mixture-sampling modes and both base models.

N is the number of $\mathbf{x}_0$ candidates drawn per denoising step — the Monte Carlo budget of the estimator. λ is the tilt in $q(\mathbf{x}_0) \propto p_\theta(\mathbf{x}_0)\, e^{\lambda r(\mathbf{x}_0)}$. The two modes differ only in how $\mathbf{x}_{t-1}$ is drawn from the resulting mixture: `marginal` averages the N individually normalised posteriors, `edlm` draws a component and then samples from it. They share per-position marginals and differ only in the joint.

2 Reading ESS: use the median, never the last step

Effective sample size $\text{ESS} = 1/\sum_n w_n^2$ says which end of the grid a run is on: $\text{ESS} = N$ means the tilt is doing nothing, $\text{ESS} \to 1$ means it has collapsed onto a single candidate.

The driver script prints only the *final* step's ESS, and that number is actively misleading on MDLM. Absorbing-state sampling has almost everything unmasked by the last step, so all N candidates decode to the same sequence and tie on reward — the last-step ESS reads $\approx N$ however hard λ is tilting. Judged on it, MDLM looks entirely unguided. Its novel QED in fact climbs across the λ grid. Every ESS below is the **median over the trajectory**, digested from the per-run logs into `results/_ess_summary.csv`.

N	$\lambda = 0$	$\lambda = 0.5$	$\lambda = 1$	$\lambda = 2$	$\lambda = 5$	$\lambda = 10$	$\lambda = 30$	$\lambda = 100$	$\lambda = 1000$
10	1.00	0.97	0.94	0.90	0.88	0.88	0.88	0.88	0.88
30	1.00	0.96	0.91	0.84	0.80	0.81	0.80	0.80	0.80
100	1.00	0.96	0.89	0.80	0.73	0.71	0.71	0.71	0.71
300	1.00	0.95	0.86	0.74	0.65	0.65	0.64	0.65	0.65
500	1.00	0.95	0.85	0.72	0.63	0.62	0.62	0.62	0.62
1000	0.00	0.94	0.83	0.70	0.59	0.58	0.58	0.58	0.58
2000	0.00	0.92	0.78	0.61	0.51	0.49	0.49	0.49	0.49

Table 1: MDLM, `marginal`: median ESS as a fraction of N.

N	$\lambda = 0$	$\lambda = 0.5$	$\lambda = 1$	$\lambda = 2$	$\lambda = 5$	$\lambda = 10$	$\lambda = 30$	$\lambda = 100$	$\lambda = 1000$
10	1.00	0.97	0.94	0.91	0.88	0.88	0.88	0.88	0.88
30	1.00	0.96	0.91	0.86	0.81	0.80	0.80	0.80	0.80
100	1.00	0.96	0.89	0.80	0.72	0.70	0.69	0.69	0.69
300	1.00	0.95	0.87	0.75	0.64	0.65	0.64	0.64	0.64
500	1.00	0.95	0.86	0.72	0.64	0.63	0.63	0.63	0.63
1000	0.00	0.94	0.82	0.68	0.60	0.59	0.58	0.58	0.58
2000	0.00	0.93	0.79	0.62	0.53	0.51	0.50	0.50	0.50

Table 2: MDLM, `edlm`: median ESS as a fraction of N.

N	$\lambda = 0$	$\lambda = 0.5$	$\lambda = 1$	$\lambda = 2$	$\lambda = 5$	$\lambda = 10$	$\lambda = 30$	$\lambda = 100$	$\lambda = 1000$
10	1.00	0.98	0.94	0.88	0.85	0.84	0.85	0.84	0.84
30	1.00	0.97	0.91	0.80	0.72	0.72	0.72	0.72	0.72
100	1.00	0.97	0.88	0.71	0.51	0.51	0.50	0.50	0.50
300	1.00	0.96	0.86	0.57	0.26	0.26	0.26	0.26	0.26
500	1.00	0.97	0.85	0.52	0.19	0.19	0.18	0.18	0.18
1000	0.00	0.97	0.85	0.43	0.11	0.11	0.11	0.11	0.11
2000	0.00	0.97	0.82	0.36	0.07	0.07	0.07	0.07	0.07

Table 3: UDLM, `marginal`: median ESS as a fraction of N.

N	$\lambda = 0$	$\lambda = 0.5$	$\lambda = 1$	$\lambda = 2$	$\lambda = 5$	$\lambda = 10$	$\lambda = 30$	$\lambda = 100$	$\lambda = 1000$
10	1.00	0.98	0.94	0.88	0.85	0.85	0.85	0.85	0.85
30	1.00	0.97	0.91	0.81	0.73	0.73	0.73	0.73	0.73
100	1.00	0.97	0.89	0.71	0.51	0.51	0.51	0.51	0.51
300	1.00	0.96	0.85	0.58	0.27	0.27	0.27	0.27	0.27
500	1.00	0.97	0.85	0.51	0.19	0.18	0.18	0.18	0.18
1000	0.00	0.96	0.83	0.42	0.11	0.11	0.11	0.11	0.11
2000	0.00	0.96	0.83	0.36	0.07	0.07	0.07	0.07	0.07

Table 4: UDLM, `edlm`: median ESS as a fraction of N.

3 The grid

N	$\lambda = 0$	$\lambda = 0.5$	$\lambda = 1$	$\lambda = 2$	$\lambda = 5$	$\lambda = 10$	$\lambda = 30$	$\lambda = 100$	$\lambda = 1000$
10	2.477	2.751	2.824	2.815	2.819	2.820	2.820	2.820	2.820
30	2.365	2.552	2.657	2.683	2.681	2.639	2.633	2.633	2.633
100	2.346	2.573	2.822	2.979	3.080	3.129	3.108	3.108	3.108
300	2.468	2.716	3.033	3.260	3.345	3.351	3.367	3.367	3.367
500	2.548	2.698	2.952	3.173	3.311	3.413	3.410	3.410	3.410
1000	2.426	2.603	2.911	3.310	3.452	3.499	3.493	3.493	3.493
2000	2.379	2.672	3.045	3.615	3.715	3.611	3.627	3.627	3.627

Table 5: MDLM, `marginal`: novel-ring count mean.

N	$\lambda = 0$	$\lambda = 0.5$	$\lambda = 1$	$\lambda = 2$	$\lambda = 5$	$\lambda = 10$	$\lambda = 30$	$\lambda = 100$	$\lambda = 1000$
10	2.567	2.707	2.810	2.864	2.783	2.773	2.770	2.770	2.770
30	2.697	2.841	2.874	2.899	2.797	2.797	2.797	2.797	2.797
100	2.284	2.550	2.854	2.834	3.039	3.107	3.103	3.103	3.103
300	2.493	2.736	2.964	3.080	3.396	3.396	3.396	3.396	3.396
500	2.318	2.724	3.107	3.146	3.243	3.194	3.184	3.184	3.184
1000	2.612	2.803	3.174	3.306	3.484	3.543	3.538	3.538	3.538
2000	2.718	2.840	3.090	3.546	3.793	3.754	3.752	3.752	3.752

Table 6: MDLM, `edlm`: novel-ring count mean.

N	$\lambda = 0$	$\lambda = 0.5$	$\lambda = 1$	$\lambda = 2$	$\lambda = 5$	$\lambda = 10$	$\lambda = 30$	$\lambda = 100$	$\lambda = 1000$
10	1.656	1.865	1.935	1.873	1.868	1.884	1.884	1.884	1.884
30	1.708	1.785	1.787	1.884	1.789	1.799	1.804	1.804	1.804
100	1.660	1.776	1.825	1.846	1.920	1.936	1.932	1.932	1.932
300	1.637	1.768	1.764	1.954	2.168	2.110	2.125	2.125	2.125
500	1.592	1.627	1.721	2.008	2.317	2.162	2.211	2.211	2.211
1000	1.543	1.646	1.842	2.129	2.302	2.364	2.351	2.351	2.351
2000	1.469	1.654	1.733	1.953	2.524	2.572	2.581	2.581	2.581

Table 7: UDLM, `marginal`: novel-ring count mean.

4 `marginal` against `edlm`

Two things were predicted before the sweep ran, and only one holds.

Predicted and confirmed: the two modes are empirically interchangeable. Across the whole grid the mean absolute gap in novel-ring count is 0.107 with mean signed gap -0.022 — no consistent winner, at any λ or N.

Predicted and *not* confirmed: the gap was expected to *shrink* as λ grows, because one-hot weights make $\sum_n w_n q(\cdot \,|\, \mathbf{x}_0^{(n)})$ collapse to the single component `edlm` would have drawn, and a posterior conditioned on a fixed $\mathbf{x}_0$ already factorises over positions. Measured, the gap is flat: 0.160 at $\lambda = 0$ against 0.096 at $\lambda = 1000$. The reason is visible in the ESS tables — the weights

N	$\lambda = 0$	$\lambda = 0.5$	$\lambda = 1$	$\lambda = 2$	$\lambda = 5$	$\lambda = 10$	$\lambda = 30$	$\lambda = 100$	$\lambda = 1000$
10	1.624	1.764	1.775	1.828	1.738	1.732	1.732	1.732	1.732
30	1.524	1.707	1.853	1.718	1.734	1.717	1.717	1.717	1.717
100	1.600	1.801	1.888	1.860	1.918	1.791	1.793	1.793	1.793
300	1.693	1.801	1.820	2.028	2.161	2.043	2.027	2.027	2.027
500	1.653	1.901	1.902	1.933	2.195	2.199	2.180	2.180	2.180
1000	1.799	1.888	1.966	2.328	2.492	2.401	2.392	2.392	2.392
2000	1.797	2.013	1.945	2.077	2.540	2.717	2.727	2.727	2.727

Table 8: UDLM, `edlm`: novel-ring count mean.

N	$\lambda = 0$	$\lambda = 0.5$	$\lambda = 1$	$\lambda = 2$	$\lambda = 5$	$\lambda = 10$	$\lambda = 30$	$\lambda = 100$	$\lambda = 1000$
10	216	233	244	243	254	256	256	256	256
30	211	232	245	265	276	288	289	289	289
100	211	232	259	281	287	279	279	279	279
300	220	236	241	281	293	316	316	316	316
500	228	245	272	294	331	334	334	334	334
1000	204	229	257	281	361	367	369	369	369
2000	190	204	245	304	354	357	362	362	362

Table 9: MDLM, `marginal`: number of novel molecules (out of 1,024 generated).

N	$\lambda = 0$	$\lambda = 0.5$	$\lambda = 1$	$\lambda = 2$	$\lambda = 5$	$\lambda = 10$	$\lambda = 30$	$\lambda = 100$	$\lambda = 1000$
10	215	242	247	243	254	256	256	256	256
30	234	258	262	258	256	251	251	251	251
100	208	242	267	289	307	309	310	310	310
300	209	231	247	275	285	288	288	288	288
500	195	214	252	294	300	319	320	320	320
1000	219	249	281	307	339	363	364	364	364
2000	213	231	266	328	367	378	379	379	379

Table 10: MDLM, `edlm`: number of novel molecules (out of 1,024 generated).

N	$\lambda = 0$	$\lambda = 0.5$	$\lambda = 1$	$\lambda = 2$	$\lambda = 5$	$\lambda = 10$	$\lambda = 30$	$\lambda = 100$	$\lambda = 1000$
10	189	208	230	251	243	241	241	241	241
30	195	214	202	242	251	264	265	265	265
100	194	214	229	240	286	267	265	265	265
300	193	207	212	240	268	292	288	288	288
500	196	217	222	249	278	284	279	279	279
1000	186	206	228	241	288	294	308	308	308
2000	207	234	243	256	275	278	277	277	277

Table 11: UDLM, `marginal`: number of novel molecules (out of 1,024 generated).

N	$\lambda = 0$	$\lambda = 0.5$	$\lambda = 1$	$\lambda = 2$	$\lambda = 5$	$\lambda = 10$	$\lambda = 30$	$\lambda = 100$	$\lambda = 1000$
10	165	182	204	233	252	257	257	257	257
30	191	222	238	266	274	276	276	276	276
100	180	211	233	265	269	296	295	295	295
300	179	191	205	251	310	301	301	301	301
500	190	222	214	252	277	297	295	295	295
1000	164	215	235	271	301	309	309	309	309
2000	212	239	253	271	313	311	319	319	319

Table 12: UDLM, `edlm`: number of novel molecules (out of 1,024 generated).

N	$\lambda = 0$	$\lambda = 0.5$	$\lambda = 1$	$\lambda = 2$	$\lambda = 5$	$\lambda = 10$	$\lambda = 30$	$\lambda = 100$	$\lambda = 1000$
10	9.000	9.000	10.000	10.000	10.000	10.000	10.000	10.000	10.000
30	8.000	10.000	8.000	8.000	8.000	8.000	8.000	8.000	8.000
100	9.000	10.000	9.000	10.000	10.000	10.000	10.000	10.000	10.000
300	7.000	14.000	14.000	14.000	14.000	14.000	14.000	14.000	14.000
500	13.000	13.000	13.000	13.000	13.000	13.000	13.000	13.000	13.000
1000	9.000	9.000	9.000	10.000	10.000	10.000	10.000	10.000	10.000
2000	8.000	10.000	10.000	12.000	12.000	12.000	12.000	12.000	12.000

A raw max grows with the number of molecules it is taken over, and these cells differ several-fold in novel count, so a higher max here is partly a deeper pool. That is a real advantage at a fixed generation budget – every cell generated the same 1,024 sequences – but it is not a claim about distribution shape. Section 7 separates the two.

Table 13: MDLM, `marginal`: highest novel-ring count of the run.

N	$\lambda = 0$	$\lambda = 0.5$	$\lambda = 1$	$\lambda = 2$	$\lambda = 5$	$\lambda = 10$	$\lambda = 30$	$\lambda = 100$	$\lambda = 1000$
10	7.000	9.000	9.000	9.000	9.000	9.000	9.000	9.000	9.000
30	9.000	10.000	10.000	10.000	10.000	10.000	10.000	10.000	10.000
100	8.000	10.000	10.000	10.000	10.000	10.000	10.000	10.000	10.000
300	8.000	10.000	10.000	10.000	9.000	9.000	9.000	9.000	9.000
500	10.000	10.000	10.000	10.000	10.000	10.000	10.000	10.000	10.000
1000	7.000	9.000	10.000	10.000	11.000	11.000	11.000	11.000	11.000
2000	8.000	10.000	10.000	10.000	10.000	10.000	10.000	10.000	10.000

Table 14: MDLM, `edlm`: highest novel-ring count of the run.

N	$\lambda = 0$	$\lambda = 0.5$	$\lambda = 1$	$\lambda = 2$	$\lambda = 5$	$\lambda = 10$	$\lambda = 30$	$\lambda = 100$	$\lambda = 1000$
10	6.000	7.000	7.000	7.000	7.000	7.000	7.000	7.000	7.000
30	6.000	7.000	7.000	7.000	6.000	6.000	6.000	6.000	6.000
100	7.000	6.000	6.000	6.000	7.000	8.000	8.000	8.000	8.000
300	5.000	7.000	8.000	7.000	7.000	6.000	6.000	6.000	6.000
500	6.000	6.000	5.000	7.000	7.000	7.000	7.000	7.000	7.000
1000	5.000	10.000	10.000	7.000	10.000	7.000	7.000	7.000	7.000
2000	5.000	6.000	5.000	7.000	6.000	10.000	10.000	10.000	10.000

Table 15: UDLM, `marginal`: highest novel-ring count of the run.

N	$\lambda = 0$	$\lambda = 0.5$	$\lambda = 1$	$\lambda = 2$	$\lambda = 5$	$\lambda = 10$	$\lambda = 30$	$\lambda = 100$	$\lambda = 1000$
10	6.000	8.000	6.000	6.000	5.000	5.000	5.000	5.000	5.000
30	7.000	7.000	7.000	7.000	7.000	7.000	7.000	7.000	7.000
100	6.000	6.000	7.000	6.000	5.000	6.000	6.000	6.000	6.000
300	7.000	7.000	7.000	8.000	8.000	7.000	7.000	7.000	7.000
500	7.000	9.000	7.000	7.000	10.000	7.000	7.000	7.000	7.000
1000	6.000	6.000	6.000	7.000	8.000	8.000	8.000	8.000	8.000
2000	7.000	7.000	6.000	7.000	10.000	10.000	10.000	10.000	10.000

Table 16: UDLM, `edlm`: highest novel-ring count of the run.

model	λ	mean $\|\Delta\|$ property	mean Δ property	mean $\|\Delta\|$ novel	mean Δ novel
MDLM	0	0.181	-0.097	15.6	-1.9
MDLM	0.5	0.110	-0.091	18.3	-8.0
MDLM	1	0.114	-0.090	14.1	-8.4
MDLM	2	0.099	+0.023	10.1	-6.4
MDLM	5	0.060	-0.019	16.3	+6.9
MDLM	10	0.097	-0.015	19.3	+4.7
MDLM	30	0.092	-0.012	19.0	+5.3
MDLM	100	0.092	-0.012	19.0	+5.3
MDLM	1000	0.092	-0.012	19.0	+5.3
UDLM	0	0.139	-0.061	12.7	+11.3
UDLM	0.5	0.159	-0.108	10.3	+2.6
UDLM	1	0.123	-0.077	14.0	-2.3
UDLM	2	0.100	-0.018	18.0	-12.9
UDLM	5	0.074	+0.016	20.4	-15.3
UDLM	10	0.095	+0.032	18.1	-18.1
UDLM	30	0.099	+0.046	18.4	-18.4
UDLM	100	0.099	+0.046	18.4	-18.4
UDLM	1000	0.099	+0.046	18.4	-18.4

A systematic winner would show as a consistent sign in the Δ columns. There is none.

Table 17: `marginal` minus `edlm`, averaged over the five N at each λ. $|\Delta|$ is the typical size of the gap, Δ its sign.

never actually become one-hot in this grid, so the limit is never entered. The flat gap is therefore sampling noise between two samplers that agree, which is the same conclusion by a weaker route.

5 $\lambda = 0$: the unguided control

At $\lambda = 0$ the weights are uniform, so no tilt is applied whatever N or the mode is. Every cell of Table 18 should therefore be the base sampler, and the spread across it measures the sampling noise that every other comparison in this report has to clear.

model	mode	$N = 10$	$N = 30$	$N = 100$	$N = 300$	$N = 500$	$N = 1000$	$N = 2000$
MDLM	marginal	2.477	2.365	2.346	2.468	2.548	2.426	2.379
MDLM	edlm	2.567	2.697	2.284	2.493	2.318	2.612	2.718
UDLM	marginal	1.656	1.708	1.660	1.637	1.592	1.543	1.469
UDLM	edlm	1.624	1.524	1.600	1.693	1.653	1.799	1.797

This is the implementation control: at $\lambda = 0$ the weights are uniform, so no tilt is applied and the numbers must be flat in N up to sampling noise. They are.

Table 18: novel-ring count at $\lambda = 0$. Every entry should be the base sampler, independent of N and of the mode.

This table also calibrates everything else in the report. The widest $\lambda = 0$ spread within a single model is 0.435 in novel-ring count — runs that are identical in distribution, differing only by sampling noise. The mean gap between `marginal` and `edlm` across the whole grid is 0.107, **smaller** than that floor. So the two modes are not merely close: their difference is indistinguishable from noise.

6 The frontier against D-CBG

Both methods are swept, then compared on the axes the paper's Figure 3 uses: number of novel molecules against novel-ring count mean. Comparing at single hyperparameter points would be meaningless — the paper tunes γ per model *and* per property.

method	setting	valid	novel	novel-ring count
ours	edlm, $N = 2000$, $\lambda = 5$	766	367	3.793
ours	edlm, $N = 2000$, $\lambda = 10$	754	378	3.754
ours	edlm, $N = 2000$, $\lambda = 30$	754	379	3.752
ours	edlm, $N = 2000$, $\lambda = 100$	754	379	3.752
ours	edlm, $N = 2000$, $\lambda = 1000$	754	379	3.752
D-CBG	$\gamma = 1$, exact	188	149	4.745
D-CBG	$\gamma = 3$, approx.	267	150	3.400
D-CBG	$\gamma = 2$, approx.	352	176	2.966
D-CBG	$\gamma = 1$, approx.	493	209	2.809

1 collapsed run(s) with fewer than 10 novel molecules were excluded; a property mean over a handful of molecules is not an operating point.

Table 19: MDLM: the Pareto-optimal settings of each method on (novel count, novel-ring count).

method	setting	valid	novel	novel-ring count
ours	edlm, $N = 2000$, $\lambda = 30$	982	319	2.727
ours	edlm, $N = 2000$, $\lambda = 100$	982	319	2.727
ours	edlm, $N = 2000$, $\lambda = 1000$	982	319	2.727
D-CBG	$\gamma = 10$, exact	892	416	4.885
D-CBG	$\gamma = 8$, exact	910	449	4.837

Table 20: UDLM: the Pareto-optimal settings of each method on (novel count, novel-ring count).

7 Beyond the mean

Everything above compares *means*, which is the paper's metric but not the one a screening pipeline faces. Nobody consumes the mean of a generated library; they take the usable molecules out of it. And the mean is blind to the axis these methods differ on most: from the same 1,024 generated sequences our runs yield several times more novel molecules than D-CBG's.

7.1 The primary comparison: usable molecules per fixed budget

Both methods generated exactly 1,024 sequences. So simply count the novel molecules that clear a quality bar. Quality and quantity both enter — a run that yields many poor novel molecules scores nothing, and one that yields a handful of excellent ones scores little.

This number needs no correction of any kind. In particular it must **not** be normalised by the novel count, nor size-matched against the competitor: converting a fixed budget into more usable molecules *is* the result, and dividing it out would erase precisely what is being measured.

7.1.1 The whole grid, on the primary metric

The tables above pick a best setting per method, which is what a headline needs but not what a reader should have to trust. Below is the raw grid: hits at the headline bar at every (N, λ), both modes, nothing averaged and nothing filtered. It shows where in the grid the hits live — and that the surface is not monotone in either axis, so a single "best" cell is partly a draw from run noise.

N	marg. 0	marg. 5	marg. 10	marg. 30	marg. 1000	edlm. 0	edlm. 5	edlm. 10	edlm. 30	edlm. 1000
10	27	46	46	46	46	36	53	53	53	53
30	28	54	56	56	56	43	53	51	51	51
100	30	74	76	75	75	27	68	74	74	74
300	33	85	92	93	93	25	82	80	80	80
500	28	93	92	92	92	23	80	82	81	81
1000	23	107	112	112	112	32	101	114	114	114
2000	26	122	107	109	109	42	125	124	125	125

Novel molecules clearing the bar, out of the 1,024 generated. Every cell is a single run, seed 1; nothing is averaged. $\lambda \in \{0.5, 1, 2, 100\}$ are omitted to fit the page. **Not** filtered by `MIN_NOVEL`, unlike the Pareto table — a collapsed run simply scores few hits here, which is the honest reading, whereas a mean over 2 molecules is not. **D-CBG's best on this bar is 76** (exact $\gamma = 1$), for comparison.

Table 21: MDLM: hits at ring count ≥ 5 across the whole (N, λ) grid. Columns are (mixture mode) $\times$ λ.

N	marg. 0	marg. 5	marg. 10	marg. 30	marg. 1000	edlm. 0	edlm. 5	edlm. 10	edlm. 30	edlm. 1000
10	4	16	15	15	15	4	8	7	7	7
30	10	8	12	12	12	6	12	11	11	11
100	5	18	16	16	16	5	7	9	9	9
300	7	21	18	19	19	8	31	17	16	16
500	4	25	26	28	28	5	19	25	24	24
1000	8	29	29	31	31	10	38	30	30	30
2000	2	29	33	34	34	7	49	54	54	54

Novel molecules clearing the bar, out of the 1,024 generated. Every cell is a single run, seed 1; nothing is averaged. $\lambda \in \{0.5, 1, 2, 100\}$ are omitted to fit the page. **Not** filtered by `MIN_NOVEL`, unlike the Pareto table — a collapsed run simply scores few hits here, which is the honest reading, whereas a mean over 2 molecules is not. **D-CBG's best on this bar is 246** (exact $\gamma = 8$), for comparison.

Table 22: UDLM: hits at ring count ≥ 5 across the whole (N, λ) grid. Columns are (mixture mode) $\times$ λ.

model	method	setting	novel	mean	≥ 4	≥ 5	≥ 6
MDLM	`marginal`	$N = 2000$, $\lambda = 5$	354	3.715	**177** (1.3×)	**122** (1.6×)	**52** (1.5×)
MDLM	`edlm`	$N = 2000$, $\lambda = 5$	367	3.793	**184** (1.3×)	**125** (1.6×)	**66** (1.9×)
MDLM	D-CBG	$\gamma = 1$, exact	149	4.745	137	76	34
UDLM	`marginal`	$N = 2000$, $\lambda = 30$	277	2.581	59 (0.1×)	34 (0.1×)	8 (0.1×)
UDLM	`edlm`	$N = 2000$, $\lambda = 10$	311	2.717	82 (0.2×)	54 (0.2×)	7 (0.1×)
UDLM	D-CBG	$\gamma = 8$, exact	449	4.837	444	246	93

No size matching and no normalisation — every run spent the same generation budget, so these are directly comparable counts.

Table 23: Novel molecules clearing each ring count bar, out of the 1,024 sequences each run generated. The two mixture-sampling modes are kept apart rather than merged into a single *ours* row, since they are different samplers. Each is at its own best setting, ranked on the middle bar; the multiplier is against D-CBG on the same model.

7.2 Secondary: shape comparisons, which discard the counts

Two standard distribution-free comparisons. Both deliberately ignore how many molecules each method produced, so neither can see the advantage above; they are here to answer a different question — whether our *distribution* is better, or only our yield.

- **Rank against rank.** Sort both sets and compare like for like: best against best, tenth percentile against tenth. The fraction of quantiles where we lead is the *quantile win rate*; 1.0 would be **first-order stochastic dominance**. Comparing the k-th *rank* directly would need equal sample sizes, so this compares at matched quantiles, the size-independent form of the same idea.
- **Draw against draw.** A_{12}, the Vargha–Delaney probability of superiority: one molecule drawn at random from each, how often is ours better? The normalised Mann–Whitney U; 0.5 is indistinguishable.

The *matched* column subsamples the larger novel set down to the smaller. It is a **decomposition, not a fairness correction** — it separates "we win because our tail is better" from "we win because our pool is deeper". The counterfactual it describes never occurs in practice, so it should not be read as the fair comparison; the table above is.

model	method	novel	mean	max	top10	top10 matched	q-win	A_{12}
MDLM	ours	304	3.615	12.000	10.100	8.814	0.43	0.39
MDLM	D-CBG	111	3.856	10.000	7.600	7.600	–	–
UDLM	ours	313	2.540	10.000	6.800	6.800	0.00	0.08
UDLM	D-CBG	416	4.885	10.000	9.100	8.575	–	–

max and *top-K* are at the fixed 1,024-sequence budget and so include our deeper novel pool. *matched* removes that, and answers only "quality or quantity?.

Table 24: Each method at its own best setting by top-10. *q-win* and A_{12} sit on the ours row and describe ours against D-CBG.

8 Validity

Validity is the axis on which the two mechanisms differ most sharply, and it is what drives the curve directions of the previous section, so it is worth reporting on its own.

Our reward sees a *clean* x_0 candidate and returns `invalid_reward` $= 0$ when RDKit cannot parse it. Raising λ therefore pushes probability towards parseable molecules before it pushes towards high ring count: an invalid candidate is treated as a ring count-zero one and suppressed. D-CBG instead perturbs the per-position logits with the gradient of a noisy classifier, which carries no such term — nothing in it prefers a parseable string.

9 The shape of each curve

Plot novel count on the x-axis against novel-ring count mean on the y-axis and sweep each method's strength dial — λ for ours, γ for D-CBG. The two methods trace curves in *opposite directions*, and

N	$\lambda = 0$	$\lambda = 0.5$	$\lambda = 1$	$\lambda = 2$	$\lambda = 5$	$\lambda = 10$	$\lambda = 30$	$\lambda = 100$	$\lambda = 1000$
10	547	580	600	610	628	629	629	629	629
30	542	569	595	623	637	637	637	637	637
100	532	568	600	638	659	657	658	658	658
300	552	580	600	657	683	700	700	700	700
500	558	584	618	650	684	696	695	695	695
1000	549	583	613	653	697	701	702	702	702
2000	544	565	611	654	716	731	734	734	734

Table 25: MDLM, `marginal`: valid molecules out of the 1,024 generated.

N	$\lambda = 0$	$\lambda = 0.5$	$\lambda = 1$	$\lambda = 2$	$\lambda = 5$	$\lambda = 10$	$\lambda = 30$	$\lambda = 100$	$\lambda = 1000$
10	564	590	605	622	629	629	629	629	629
30	558	590	608	622	639	640	640	640	640
100	553	598	631	656	704	699	698	698	698
300	556	594	620	659	680	696	696	696	696
500	549	581	614	658	696	704	704	704	704
1000	542	577	631	678	717	728	730	730	730
2000	559	591	633	694	766	754	754	754	754

Table 26: MDLM, `edlm`: valid molecules out of the 1,024 generated.

N	$\lambda = 0$	$\lambda = 0.5$	$\lambda = 1$	$\lambda = 2$	$\lambda = 5$	$\lambda = 10$	$\lambda = 30$	$\lambda = 100$	$\lambda = 1000$
10	682	731	770	832	858	866	867	867	867
30	695	759	791	867	902	922	922	922	922
100	686	743	798	863	916	921	921	921	921
300	672	733	774	872	940	957	961	961	961
500	691	744	792	862	943	953	951	951	951
1000	695	748	792	880	938	954	961	961	961
2000	694	763	817	873	954	972	974	974	974

Table 27: UDLM, `marginal`: valid molecules out of the 1,024 generated.

N	$\lambda = 0$	$\lambda = 0.5$	$\lambda = 1$	$\lambda = 2$	$\lambda = 5$	$\lambda = 10$	$\lambda = 30$	$\lambda = 100$	$\lambda = 1000$
10	657	731	786	845	889	890	890	890	890
30	675	744	802	879	912	919	919	919	919
100	679	746	813	879	944	953	953	953	953
300	693	765	810	901	969	978	979	979	979
500	686	775	806	882	966	979	980	980	980
1000	685	748	808	901	978	984	986	986	986
2000	685	766	824	902	981	979	982	982	982

Table 28: UDLM, `edlm`: valid molecules out of the 1,024 generated.

model	method	weakest dial	best	strongest dial	trend
MDLM	ours ($N = 2000$)	544	734	734	**rises**
MDLM	D-CBG exact	188	188	2	falls
MDLM	D-CBG approx.	493	493	88	falls
UDLM	ours ($N = 2000$)	694	974	974	**rises**
UDLM	D-CBG exact	807	910	892	rises
UDLM	D-CBG approx.	992	992	792	falls

This is the mechanism behind the opposite curve directions: our dial buys validity on the way to buying property, so the novel count rises with it, while D-CBG spends validity and the novel count falls. On MDLM the D-CBG exact arm reaches zero valid molecules at $\gamma \geq 6$, which is why the paper does not report it there.

Table 29: Valid molecules out of 1,024 as the strength dial is turned up. Ours is read along λ at the largest N; D-CBG along γ.

that is a property worth reporting on its own, independently of which sits higher.

- **D-CBG runs up and to the left.** Raising γ buys property by spending novelty: the dial is a genuine trade-off the user has to tune, and the original paper tunes it per model and per property.
- **Ours runs up and to the right.** Raising λ improves *both* axes at once. The reason is visible in the validity column: the first thing λ buys is valid molecules, which raises the novel count, and only then does it buy property. There is no trade-off to tune — the only cost is compute.

The correlation between the two axes along each sweep reduces this to one number. Crucially it is measured at *fixed* N, so the up-right shape is not an artifact of spending more Monte Carlo budget: along every row below, only the strength dial moves.

model	method	pts	weakest dial	strongest dial	corr	shape
MDLM	ours, $N = 100$	9	(211, 2.346)	(279, 3.108)	+0.98	**up-right**
MDLM	ours, $N = 300$	9	(220, 2.468)	(316, 3.367)	+0.93	**up-right**
MDLM	ours, $N = 1000$	9	(204, 2.426)	(369, 3.493)	+0.96	**up-right**
MDLM	ours, $N = 2000$	9	(190, 2.379)	(362, 3.627)	+0.96	**up-right**
MDLM	D-CBG approx.	6	(209, 2.809)	(67, 4.597)	-0.99	up-left
UDLM	ours, $N = 100$	9	(194, 1.660)	(265, 1.932)	+0.95	**up-right**
UDLM	ours, $N = 300$	9	(193, 1.637)	(288, 2.125)	+0.96	**up-right**
UDLM	ours, $N = 1000$	9	(186, 1.543)	(308, 2.351)	+0.97	**up-right**
UDLM	ours, $N = 2000$	9	(207, 1.469)	(277, 2.581)	+0.96	**up-right**
UDLM	D-CBG exact	6	(275, 4.033)	(416, 4.885)	+0.86	**up-right**
UDLM	D-CBG approx.	6	(154, 2.519)	(355, 4.780)	+0.87	**up-right**

Measured at fixed N, so the up-right shape is not an effect of raising the Monte Carlo budget — only the strength dial moves along each row.

Table 30: Direction of each curve in (novel count, novel-ring count) space. *corr* is between the two axes along the sweep: positive means the dial improves both, negative means it trades one for the other.

ours	$\lambda = 0$	$\lambda = 0.5$	$\lambda = 1$	$\lambda = 2$	$\lambda = 5$	$\lambda = 10$	$\lambda = 30$	$\lambda = 100$	$\lambda = 1000$
$N = 10$	(216, 2.477)	(233, 2.751)	(244, 2.824)	(243, 2.815)	(254, 2.819)	(256, 2.820)	(256, 2.820)	(256, 2.820)	(256, 2.820)
$N = 30$	(211, 2.365)	(232, 2.552)	(245, 2.657)	(265, 2.683)	(276, 2.681)	(288, 2.639)	(289, 2.633)	(289, 2.633)	(289, 2.633)
$N = 100$	(211, 2.346)	(232, 2.573)	(259, 2.822)	(281, 2.979)	(287, 3.080)	(279, 3.129)	(279, 3.108)	(279, 3.108)	(279, 3.108)
$N = 300$	(220, 2.468)	(236, 2.716)	(241, 3.033)	(281, 3.260)	(293, 3.345)	(316, 3.351)	(316, 3.367)	(316, 3.367)	(316, 3.367)
$N = 500$	(228, 2.548)	(245, 2.698)	(272, 2.952)	(294, 3.173)	(331, 3.311)	(334, 3.413)	(334, 3.410)	(334, 3.410)	(334, 3.410)
$N = 1000$	(204, 2.426)	(229, 2.603)	(257, 2.911)	(281, 3.310)	(361, 3.452)	(367, 3.499)	(369, 3.493)	(369, 3.493)	(369, 3.493)
$N = 2000$	(190, 2.379)	(204, 2.672)	(245, 3.045)	(304, 3.615)	(354, 3.715)	(357, 3.611)	(362, 3.627)	(362, 3.627)	(362, 3.627)

Table 31: MDLM: the (novel, novel-ring count) point at each λ, for every N. Reading a row left to right traces that curve.

D-CBG						
exact	$\gamma = 1$: (149, 4.745)					
approx.	$\gamma = 1$: (209, 2.809)	$\gamma = 2$: (176, 2.966)	$\gamma = 3$: (150, 3.400)	$\gamma = 5$: (111, 3.856)	$\gamma = 8$: (86, 4.244)	$\gamma = 10$: (67, 4.597)

Table 32: MDLM: the same for D-CBG, sweeping γ. The novel count falls as the property rises — the opposite direction.

ours	$\lambda = 0$	$\lambda = 0.5$	$\lambda = 1$	$\lambda = 2$	$\lambda = 5$	$\lambda = 10$	$\lambda = 30$	$\lambda = 100$	$\lambda = 1000$
$N = 10$	(189, 1.656)	(208, 1.865)	(230, 1.935)	(251, 1.873)	(243, 1.868)	(241, 1.884)	(241, 1.884)	(241, 1.884)	(241, 1.884)
$N = 30$	(195, 1.708)	(214, 1.785)	(202, 1.787)	(242, 1.884)	(251, 1.789)	(264, 1.799)	(265, 1.804)	(265, 1.804)	(265, 1.804)
$N = 100$	(194, 1.660)	(214, 1.776)	(229, 1.825)	(240, 1.846)	(286, 1.920)	(267, 1.936)	(265, 1.932)	(265, 1.932)	(265, 1.932)
$N = 300$	(193, 1.637)	(207, 1.768)	(212, 1.764)	(240, 1.954)	(268, 2.168)	(292, 2.110)	(288, 2.125)	(288, 2.125)	(288, 2.125)
$N = 500$	(196, 1.592)	(217, 1.627)	(222, 1.721)	(249, 2.008)	(278, 2.317)	(284, 2.162)	(279, 2.211)	(279, 2.211)	(279, 2.211)
$N = 1000$	(186, 1.543)	(206, 1.646)	(228, 1.842)	(241, 2.129)	(288, 2.302)	(294, 2.364)	(308, 2.351)	(308, 2.351)	(308, 2.351)
$N = 2000$	(207, 1.469)	(234, 1.654)	(243, 1.733)	(256, 1.953)	(275, 2.524)	(278, 2.572)	(277, 2.581)	(277, 2.581)	(277, 2.581)

Table 33: UDLM: the (novel, novel-ring count) point at each λ, for every N. Reading a row left to right traces that curve.

D-CBG						
exact	$\gamma = 1$: (275, 4.033)	$\gamma = 2$: (307, 4.619)	$\gamma = 3$: (345, 4.557)	$\gamma = 5$: (400, 4.695)	$\gamma = 8$: (449, 4.837)	$\gamma = 10$: (416, 4.885)
approx.	$\gamma = 1$: (154, 2.519)	$\gamma = 2$: (147, 3.184)	$\gamma = 3$: (197, 3.893)	$\gamma = 5$: (223, 4.435)	$\gamma = 8$: (298, 4.661)	$\gamma = 10$: (355, 4.780)

Table 34: UDLM: the same for D-CBG, sweeping γ. The novel count falls as the property rises — the opposite direction.

10 What the sweep says

MDLM. Peak novel-ring count: ours 3.793 against D-CBG 4.745. Peak novel count: ours 379 against D-CBG 209. Each method wins one axis, so the comparison is a genuine frontier. **UDLM.** Peak novel-ring count: ours 2.727 against D-CBG 4.885. Peak novel count: ours 319 against D-CBG 449. *D-CBG dominates on both axes* — there is no trade-off to report here, we simply lose.

10.1 Why ring count behaves differently from QED

The method can only *select*, never *invent*. Weights are softmax(λr) over N candidates drawn from x_θ, so the best reachable reward at a step is the best value *present among those N candidates*. No λ can produce a 5-ring molecule if none of the N proposals has one.

That ceiling binds hard on ring count and barely at all on QED, for two reasons that reinforce each other:

- **The reward is integer valued**, so candidates tie and the tilt saturates early. Raising λ past ≈ 5 changes nothing: the softmax is already a hard argmax over the tied top set.
- **High-ring molecules are rare under the base model** (QM9's mean ring count is 1.74), so the N proposals rarely contain one.

D-CBG has no such ceiling: it perturbs the per-position logits with a classifier gradient and can steer toward regions the base model would not have proposed. That is exactly what the numbers show.

The binding constraint is N, not λ. Along the saturated λ ridge the property still climbs monotonically with N and has not flattened at $N = 500$. Pushing N further is the one lever with headroom left; raising λ is spent.